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Physics-Informed Neural Network Solution of the 2D Helmholtz Equation with a Gaussian Source

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ABSTRACT

We present a physics-informed neural network (PINN) framework for solving the complex-valued two-dimensional Helmholtz equation with a localized Gaussian source and spatially varying permittivity. Starting from Maxwell's equations, the frequency-domain scalar Helmholtz formulation under transverse electric (TE) polarization is derived and enforced directly within the neural network loss function. The model employs a sinusoidal representation network (SIREN) architecture to capture the oscillatory nature of wave solutions and incorporates the Sommerfeld radiation condition to impose open boundaries. Training is performed using a hybrid collocation strategy combined with a two-stage optimization procedure (Adam followed by L-BFGS). Numerical experiments in free space and in the presence of a dielectric inclusion demonstrate physically consistent wave propagation, refraction, and scattering behavior. The resulting PINN achieves low residual error without requiring labeled simulation data and serves as a mesh-free surrogate capable of rapid field evaluation after training. These results highlight the potential of physics-informed neural networks for frequency-domain electromagnetic simulation and design acceleration.

INDEX TERMS

Physics-informed neural networks, Helmholtz equation, computational electromagnetics, SIREN, multiphysics modeling.

I. INTRODUCTION

Electromagnetic wave behavior in complex materials is described by Maxwell's equations [1], [2]. Under time-harmonic conditions, when wave oscillations are steady over time, these equations reduce to the Helmholtz equation. Such formulations are central to a wide range of physical systems, including wave propagation in structured media and the analysis of material-dependent electromagnetic responses.

In this project, the objective is to solve the two-dimensional Helmholtz equation in the presence of a localized Gaussian

source with standard deviation $\sigma = 0.05$ and spatially varying permittivity.

Rather than relying on traditional numerical techniques such as finite-difference or finite-element methods, which require spatial discretization and can be computationally demanding, we adopt a Physics-Informed Neural Network (PINN) approach [3], [4]. PINNs provide a mesh-free, data-efficient alternative by training a neural network to approximate a solution that satisfies the governing physics. This paradigm has seen increasing adoption across a broad range of scientific domains, including wave physics, electronic structure modeling, complex systems analysis, and data-driven approaches to multiphysics simulation [5]–[9].

The main objective of this project is to solve the two-dimensional Helmholtz equation, at a normalized angular frequency $\omega = 12.0$, arising from electromagnetic wave propagation in complex media using a Physics-Informed Neural Network (PINN). PINNs have recently been applied to Helmholtz-type and wave propagation problems, including recent formulations involving localized Gaussian sources [10]. The approach begins with deriving the governing equation from Maxwell's equations. Following this, a PINN is implemented using the SIREN (Sinusoidal Representation Networks) architecture. Finally, the resulting solution is analyzed to verify its physical consistency.

Beyond its theoretical interest, this setup is representative of practical scenarios in antenna design, integrated photonics, and metasurface engineering, where many frequency-domain simulations are required while exploring different geometries and materials. Accurate modeling of underlying material properties, including electronic structure and interfacial effects, remains essential in many of these applications [5]. In parallel, concepts from statistical physics and entropy-based analysis have been employed to understand complex system behavior across domains, offering complementary perspectives on emergent dynamics [6]. Advances in first-principles modeling have provided detailed insight into interfacial phenomena, including adsorption and stabilization processes in electrochemical environments [11], as well as atomistic mechanisms governing material growth and phase transitions at surfaces [12]. Furthermore, studies of thin film growth and collective diffusion processes highlight the importance of mesoscale material dynamics, including temperature-dependent diffusion and rapid mass transport phenomena, in determining emergent

structural and physical properties [13], [14]. These developments, together with broader multiscale simulation efforts in coupled physical and chemical systems [8], [9], emphasize the need for efficient and flexible computational frameworks. Traditional solvers (FEM/FDTD) are accurate but computationally expensive [15], [16], creating a bottleneck in design-space exploration. A PINN-based surrogate, as developed here, aims to alleviate this by providing a mesh-free, differentiable model of the field that can be evaluated rapidly once trained, enabling faster design-test-iterate cycles.

II. DERIVING THE HELMHOLTZ EQUATION FROM MAXWELL'S EQUATIONS

The derivation begins with Maxwell's equations in a source-driven, linear, isotropic, and non-magnetic medium:

Gauss's Law

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon} \quad (1)$$

Gauss's Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0 \quad (2)$$

Faraday's Law

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t} \quad (3)$$

Ampère–Maxwell Law

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} + \mathbf{J} \quad (4)$$

By taking the curl of Faraday's law (3), we get:

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= \nabla \times \left(-\mu_0 \frac{\partial \mathbf{H}}{\partial t} \right) \Rightarrow \\ \nabla \times (\nabla \times \mathbf{E}) &= -\mu_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) \end{aligned} \quad (5)$$

Substituting Ampère's Law (4) into the equation above (5), we obtain:

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= -\mu_0 \frac{\partial}{\partial t} \left(\epsilon \frac{\partial \mathbf{E}}{\partial t} + \mathbf{J} \right) \Rightarrow \\ \nabla \times (\nabla \times \mathbf{E}) &= -\mu_0 \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} - \mu_0 \frac{\partial \mathbf{J}}{\partial t} \end{aligned} \quad (6)$$

Applying the vector identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \quad (7)$$

For a charge-free region, we have $\nabla \cdot \mathbf{E} = 0$, so the identity simplifies to:

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \\ \nabla \times (\nabla \times \mathbf{E}) &= -\nabla^2 \mathbf{E} \end{aligned} \quad (8)$$

Substituting (8) into (6), the time-domain wave equation is obtained:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial \mathbf{J}}{\partial t} \quad (9)$$

Assume a time-harmonic electric field expressed as:

$$\mathbf{E}(\mathbf{r}, t) = \Re \{ \mathbf{E}(\mathbf{r}) e^{-i\omega t} \} \quad (10)$$

where $\mathbf{E}(\mathbf{r})$ is the complex spatial phasor. This field is real-valued and represents the measurable physical quantity.

We now compute the first and second time derivatives of $\mathbf{E}(\mathbf{r}, t)$:

$$\frac{\partial \mathbf{E}(\mathbf{r}, t)}{\partial t} = \Re \left\{ \frac{\partial}{\partial t} [\mathbf{E}(\mathbf{r}) e^{-i\omega t}] \right\} \quad (11)$$

$$= \Re \{ -i\omega \mathbf{E}(\mathbf{r}) e^{-i\omega t} \} \quad (12)$$

$$\frac{\partial^2 \mathbf{E}(\mathbf{r}, t)}{\partial t^2} = \Re \left\{ \frac{\partial^2}{\partial t^2} [\mathbf{E}(\mathbf{r}) e^{-i\omega t}] \right\} \quad (13)$$

$$= \Re \{ (-i\omega)^2 \mathbf{E}(\mathbf{r}) e^{-i\omega t} \} \quad (14)$$

$$= \Re \{ -\omega^2 \mathbf{E}(\mathbf{r}) e^{-i\omega t} \} \quad (15)$$

Both time derivatives of the real field correspond to taking derivatives of the complex exponential, and then taking the real part of the result.

To simplify calculations, we work directly with the complex field $\mathbf{E}(\mathbf{r}) e^{-i\omega t}$ and apply differential operators to it. This is valid due to the linearity of Maxwell's equations and differential operators. After solving the equations in the frequency domain, the physical, real-valued field is recovered by taking the real part:

$$\mathbf{E}_{\text{real}}(\mathbf{r}, t) = \Re \{ \mathbf{E}(\mathbf{r}) e^{-i\omega t} \} \quad (16)$$

This approach encodes both amplitude and phase information and allows the time derivatives to be replaced algebraically by:

$$\frac{\partial}{\partial t} \longrightarrow -i\omega, \quad \frac{\partial^2}{\partial t^2} \longrightarrow -\omega^2 \quad (17)$$

within the complex domain.

Therefore equation (9) takes the following form

$$\nabla^2 \mathbf{E} + \omega^2 \mu_0 \epsilon \mathbf{E} = -i\omega \mu_0 \mathbf{J} \quad (18)$$

In this work, we adopt normalized units with $\mu_0 = 1$. Under this choice, the frequency-domain Helmholtz equation becomes:

$$\nabla^2 \mathbf{E} + \omega^2 \epsilon \mathbf{E} = -i\omega \mathbf{J} \quad (19)$$

This is the form directly implemented in the Physics-Informed Neural Network solver.

III. TRANSVERSE ELECTRIC (TE) POLARIZATION AND 2D REDUCTION

The vector wave equation derived in the previous section (18) can be further simplified under the assumption of *Transverse Electric (TE)* polarization in a two-dimensional setting. In this configuration, the fields are assumed to be invariant along the z -axis:

$$\frac{\partial}{\partial z} = 0 \quad (20)$$

This assumption is appropriate for planar structures such as waveguides or metasurfaces, where wave propagation occurs in the x - y plane and the structure is uniform in z .

In TE polarization, by definition, the electric field has no components in the direction of propagation (i.e., in-plane), and only the out-of-plane component E_z is non-zero:

$$\mathbf{E}(x, y) = E_z(x, y) \hat{z}, \quad \mathbf{H}(x, y) = H_x(x, y) \hat{x} + H_y(x, y) \hat{y} \quad (21)$$

Similarly, the impressed source current is assumed to be polarized in the same direction:

$$\mathbf{J}(x, y) = J_z(x, y) \hat{z} \quad (22)$$

Substituting equations (21) and (22) into the vector Helmholtz equation:

$$\nabla^2 \mathbf{E} + \omega^2 \mu_0 \epsilon(x, y) \mathbf{E} = -i\omega \mu_0 \mathbf{J} \quad (23)$$

and recognizing that only the z -components are non-zero, we obtain the scalar Helmholtz equation for E_z :

$$\nabla^2 E_z(x, y) + \omega^2 \mu_0 \epsilon(x, y) E_z(x, y) = -i\omega \mu_0 J_z(x, y) \quad (24)$$

This equation governs the spatial behavior of the out-of-plane electric field E_z in the presence of a source term J_z and spatially varying relative permittivity $\epsilon(x, y)$.

Finally, under a normalized unit system where the magnetic permeability is set to $\mu_0 = 1$, the equation simplifies to:

$$\nabla^2 E_z(x, y) + \omega^2 \epsilon(x, y) E_z(x, y) = -i\omega J_z(x, y) \quad (25)$$

We impose an open boundary via the Sommerfeld radiation condition [17], [18]:

$$\left(\frac{\partial}{\partial n} + ik \right) E_z = 0, \quad (26)$$

applied on the square boundary of the domain, with $k = \omega\sqrt{\epsilon}$. In our setup the boundary lies in free space, so $\epsilon = 1$ and $k = \omega$.

Equation (25) is the fundamental PDE that we aim to solve using a Physics-Informed Neural Network (PINN). The neural network is trained to approximate the complex-valued function $E_z(x, y)$ such that the residual of this equation is minimized across the computational domain.

IV. NEURAL NETWORK ARCHITECTURE

The Physics-Informed Neural Network is constructed as a fully connected feed-forward neural network that approximates the solution of the scalar Helmholtz equation:

$$\nabla^2 E_z(x, y) + \omega^2 \epsilon(x, y) E_z(x, y) = -i\omega J_z(x, y) \quad (27)$$

We define the model as a mapping:

$$f_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad f_\theta(x, y) = [E_r(x, y), E_i(x, y)] \quad (28)$$

where θ denotes the network parameters, and the complex field is expressed as:

$$E_z(x, y) = E_r(x, y) + iE_i(x, y) \quad (29)$$

The network takes as input spatial coordinates (x, y) , and outputs the real and imaginary components of the electric field. Each hidden layer applies a periodic activation function.

We use sinusoidal activation functions as proposed in the SIREN (Sinusoidal Representation Networks) architecture:

$$\sigma(z) = \sin(\omega_0 z) \quad (30)$$

where $\omega_0 \in \mathbb{R}$ is a hyperparameter that controls the frequency of the activations in the first layer.

This choice is motivated by the oscillatory nature of wave equations: the solutions to the Helmholtz equation are typically superpositions of trigonometric functions, and thus:

$$E_z(x, y) \sim \sum_k A_k \sin(\mathbf{k} \cdot \mathbf{r} + \phi_k) \quad (31)$$

where $\mathbf{r} = (x, y)$, and A_k, ϕ_k are amplitude and phase terms. The sine activation allows the network to naturally represent such high-frequency components.

An important property of the sine function is that it remains stable under differentiation:

$$\frac{d}{dz} \sin(\omega_0 z) = \omega_0 \cos(\omega_0 z), \quad \frac{d^2}{dz^2} \sin(\omega_0 z) = -\omega_0^2 \sin(\omega_0 z) \quad (32)$$

This is crucial for PINNs, where second-order derivatives like $\nabla^2 E_z$ are computed via automatic differentiation. The use of smooth periodic functions leads to numerically stable gradients during training.

The neural network used in this work consists of four hidden layers with 128 neurons each, using sinusoidal activations as in the SIREN framework. This architecture was chosen to provide sufficient expressiveness to capture the oscillatory behavior inherent in solutions to the Helmholtz equation. The number of layers and neurons was selected through a combination of theoretical considerations and empirical validation. In particular, wave-like PDEs such as the Helmholtz and Schrödinger equations are known to require high-capacity models capable of resolving fine-scale features. As shown in the original SIREN paper by Sitzmann et al. [19], sinusoidal networks with 4 to 6 hidden layers achieve state-of-the-art results in learning implicit representations of complex, high-frequency signals. Our experiments confirmed that using four layers was sufficient to model the field structure accurately while maintaining stable convergence.

No explicit weight regularization (such as L_1 or L_2 penalties) was applied during training. This is a common practice in PINNs, since the network is not trained on noisy or overfitted data, but rather on satisfying a PDE that encodes the governing physics. In this case, the residual of the Helmholtz equation itself acts as a form of regularization, promoting smoothness and physical consistency in the solution. The smooth nature of the sine activation functions also inherently discourages overfitting or oscillatory artifacts.

Training was carried out in two phases. First, Adam [20] with initial learning rate 10^{-3} , a cosine annealing schedule, and gradient clipping was used for up to 20,000 iterations. Then, a limited-memory BFGS (LBFGS) [21] refinement for several hundred iterations was applied to further reduce the residual. Each training step of the PINN consists of a full forward and backward pass over all collocation points in the spatial domain. Specifically, the network takes as input the coordinates (x, y) and predicts the corresponding values of

the real and imaginary parts of the electric field, $E_r(x, y)$ and $E_i(x, y)$, respectively. Using automatic differentiation, the second-order derivatives $\nabla^2 E_r$ and $\nabla^2 E_i$ are computed directly from the network output. These derivatives are then used to evaluate the residual of the complex-valued Helmholtz equation. The residuals at each point are aggregated into a scalar loss value using the mean squared error formulation. Finally, a single optimization step (e.g., using the Adam optimizer) is performed to update the network parameters in a direction that reduces the total residual loss. This process is repeated for a fixed number of steps, each representing a full pass over the entire training domain.

Adam is widely used in PINN literature due to its ability to adapt learning rates per parameter, which helps stabilize training when computing higher-order derivatives through automatic differentiation.

The training objective is to minimize a physics-based residual across the domain. More precisely, let $\Omega \subset \mathbb{R}^2$ denote the interior domain and $\partial\Omega$ its boundary. Writing the complex field as $E_z = E_r + iE_i$, the interior residuals are defined as

$$\mathcal{R}_r = \nabla^2 E_r + \omega^2 \epsilon(x, y) E_r + \omega J_i, \quad (33)$$

$$\mathcal{R}_i = \nabla^2 E_i + \omega^2 \epsilon(x, y) E_i - \omega J_r, \quad (34)$$

where $J_z = J_r + iJ_i$ (in this work $J_r = J_z$, $J_i = 0$).

The interior loss is given by

$$\mathcal{L}_{\text{int}} = \frac{1}{N_{\text{int}}} \sum_{k=1}^{N_{\text{int}}} (\mathcal{R}_r^2(x_k, y_k) + \mathcal{R}_i^2(x_k, y_k)). \quad (35)$$

The boundary residual corresponding to the Sommerfeld condition is

$$\mathcal{R}_{\text{bc}} = \frac{\partial E_z}{\partial n} + ikE_z, \quad (36)$$

with associated boundary loss

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{k=1}^{N_{\text{bc}}} |\mathcal{R}_{\text{bc}}(x_k, y_k)|^2. \quad (37)$$

The total loss minimized during training is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{int}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} \quad (38)$$

where λ_{bc} balances interior and boundary enforcement.

While it is possible to define an automatic early-stopping rule, in practice a fixed number of iterations was used, and convergence was assessed by monitoring the residual decay and visually inspecting the fields to confirm that the solution stabilized at a low-residual, physically consistent configuration.

During training, collocation points are resampled each epoch: $N_{\text{interior}} = 40,000$ interior points (with a fraction drawn from a narrow Gaussian around the source at $(0.5, 0.0)$) and $N_{\text{boundary}} = 6,000$ boundary points on the square $[-1, 1]^2$. A uniform grid is used only for evaluation/visualization after training. The number 128 was chosen to balance numerical resolution and computational efficiency. A higher resolution

(e.g., 256 points per axis) would increase accuracy but also require more memory and training time, while a lower resolution (e.g., 64) may fail to capture fine-scale oscillatory features of the solution. With 128 points along each axis, the grid spacing is approximately $\Delta x = \Delta y = 0.0157$, which provides sufficient coverage to enforce the PDE residual accurately across the domain. Full-batch training was feasible given the low dimensionality and modest domain size. In more complex geometries or in three-dimensional settings, mini-batch training or adaptive sampling strategies could be employed to improve efficiency and convergence.

For all simulations in this work, the angular frequency was set to $\omega = 12$, corresponding to several wavelengths between the source and the boundaries.

V. IMPLEMENTATION

The Physics-Informed Neural Network (PINN) was implemented in `PyTorch`, using automatic differentiation to compute the derivatives required for the Helmholtz residual. The computational domain is the square region $[-1, 1] \times [-1, 1]$. During training, collocation points are *resampled at each epoch*: most points are drawn uniformly in the interior and on the boundary, and a fixed fraction are sampled from a narrow Gaussian around the source to better resolve the near field.

In the absence of a fixed labeled dataset, this collocation strategy plays the role of data acquisition and preparation. Initial experiments with purely uniform sampling showed that the model under-resolved the strong gradients near the source and converged slowly. To address this, a hybrid sampling scheme was adopted. A large set of points is drawn uniformly over the square domain to maintain global coverage, while a significant fraction is drawn from a narrow Gaussian centered at the source, effectively performing importance sampling in the most challenging region. Boundary points are sampled uniformly along all four edges with analytically defined outward normals so that the Sommerfeld radiation condition can be enforced consistently. Collocation points are regenerated at every epoch, which acts as a continuous data refresh mechanism and reduces any bias toward a particular discrete grid.

Developing a robust training procedure required iterating on both this sampling strategy and the optimization method. Early runs with only uniform sampling and Adam optimization produced qualitatively correct but blurry fields and relatively high residuals, indicating that neither the data distribution nor the optimizer alone was sufficient. Incorporating the source-focused sampling significantly improved the near-field accuracy, but the training loss still plateaued. Motivated by results from the PINN literature, we adopt a two-stage optimization scheme. Adam with a cosine-annealed learning rate is first used to explore the loss landscape and obtain a suitable initial solution, after which an L-BFGS refinement phase leverages curvature information to further minimize the physics-based residual. This combination proved crucial for driving down the Helmholtz residual and sharpening the predicted fields, and highlighted the practical lesson that, in PINNs, both how we sample the domain and how we optimize the network must be tuned together.

The relative permittivity $\epsilon(x, y)$ is defined according to the simulation case. In the free-space case, $\epsilon(x, y) = 1$ everywhere, while in the dielectric-circle case, $\epsilon(x, y) = 2$ inside a circle of radius $r = 0.3$ centered at the origin and $\epsilon(x, y) = 1$ elsewhere. The impressed source current $J_z(x, y)$ is modeled as a Gaussian distribution centered at $(x_0, y_0) = (0.5, 0.0)$:

$$J_z(x, y) = \exp\left(-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2}\right), \quad (39)$$

with $\sigma = 0.05$. The angular frequency is fixed at $\omega = 12$ for all simulations.

The network maps spatial coordinates (x, y) to the real and imaginary parts of the electric field (E_r, E_i) , following the SIREN architecture. It consists of an input layer with two neurons, *four* hidden layers with 128 neurons each using sinusoidal activations $\sin(\omega_0 z)$, and an output layer with two neurons representing (E_r, E_i) with linear activation. The SIREN weight initialization scheme is used to preserve stability when representing high-frequency oscillations.

The loss function is the mean squared residual of the complex Helmholtz equation

$$\nabla^2 E_z + \omega^2 \epsilon(x, y) E_z = -i\omega J_z(x, y), \quad (40)$$

evaluated at stochastically sampled collocation points in the interior. Writing $E_z = E_r + iE_i$ and separating into real and imaginary parts yields two coupled scalar PDEs. Automatic differentiation computes the required second-order derivatives, which are used to form the Laplacian in the residuals. The total interior loss is the mean squared sum of the real and imaginary residuals.

An open boundary is enforced using the Sommerfeld radiation condition:

$$\left(\frac{\partial}{\partial n} + ik\right) E_z = 0, \quad (41)$$

applied along uniformly sampled boundary points, where $k = \omega\sqrt{\epsilon}$. Since the boundary lies in free space, $\epsilon = 1$ and $k = \omega$. The normal derivative $\partial/\partial n$ is computed via automatic differentiation, and the boundary residual is weighted and added to the total loss.

Training is performed in two phases. First, the Adam optimizer is used with an initial learning rate of 10^{-3} , a cosine annealing schedule, and gradient clipping for up to 20,000 iterations. Then, a limited-memory BFGS (LBFGS) optimization is applied as a refinement step (several hundred iterations) to reduce the residual further. After training, the network is evaluated on a uniform grid for visualization; we plot the real and imaginary parts to analyze the field distribution.

VI. RESULTS AND DISCUSSION

We present PINN solutions to the 2D Helmholtz problem in two scenarios. The first considers a Gaussian source in free space, and the second considers the same source in the presence of a dielectric circle of relative permittivity $\epsilon = 2$ (radius $r = 0.3$) centered at the origin. In all runs we used $\omega = 12$ and a Gaussian source width $\sigma = 0.05$, with

the Sommerfeld radiation condition enforced on the square boundary.

The computational domain is defined as the square region $[-1, 1] \times [-1, 1]$ in normalized units, with x and y denoting spatial coordinates in the simulation plane. The source is placed at $(0.5, 0.0)$, and all distances are expressed in this normalized coordinate system. The choice of $[-1, 1]$ is purely a modeling convenience and corresponds to a finite physical size after normalization, enabling the imposition of boundary conditions at a fixed distance from the source.

The color scale in the plots represents the real part of the complex electric field $E_z(x, y)$. Since E_z is a complex quantity, its real part can take both positive and negative values. Positive values (typically shown in red) indicate that the field is in phase with the source, while negative values (shown in blue) indicate a phase shift of approximately π (out-of-phase). This sign change is a direct consequence of the oscillatory nature of solutions to the Helmholtz equation and is not an error. If the magnitude $|E_z|$ were plotted instead, the values would be strictly non-negative, but phase information would be lost.

In free space (Fig. 1), the field behaves like an outgoing 2D cylindrical wave excited by the localized source. The solution exhibits circular symmetry, evenly spaced phase fronts, and $\Re\{E_z\}/\Im\{E_z\}$ in quadrature. Introducing the dielectric circle (Fig. 2) perturbs this symmetry. Inside the inclusion, the increased ϵ reduces phase velocity so phase fronts appear slightly closer together, while at the boundary partial reflection and refraction generate subtle interference fringes. These patterns are visible in both the real and imaginary components and align with the governing physics.

The colorbars in (c,d) are symmetric about zero, so equal-intensity red and blue correspond to equal-magnitude positive and negative field values. This allows direct visual comparison of sign and phase behavior across both cases.

To quantify accuracy, the trained model is evaluated on a uniform grid and the root-mean-square residual of the complex Helmholtz equation is computed. Since the objective is to satisfy the governing PDE rather than match labeled data, this residual serves as the primary performance metric. In the final configuration, the combination of hybrid collocation sampling and two-stage Adam plus L-BFGS optimization yields a residual on the order of 10^{-4} , consistent with the physically correct wave patterns observed in both the free-space and dielectric scenarios.

Figure 3 illustrates convergence behavior. The early decay corresponds to learning the dominant wave structure, while the later phase refines higher-frequency components and improves boundary enforcement. The additional reduction during L-BFGS confirms the benefit of second-order optimization for the Helmholtz equation.

A trade-off arises in balancing the interior PDE residual and the boundary residual through the Sommerfeld weighting parameter. If the weight is too small, boundary reflections appear; if too large, interior accuracy degrades. The selected value suppresses artificial reflections while maintaining low interior error.

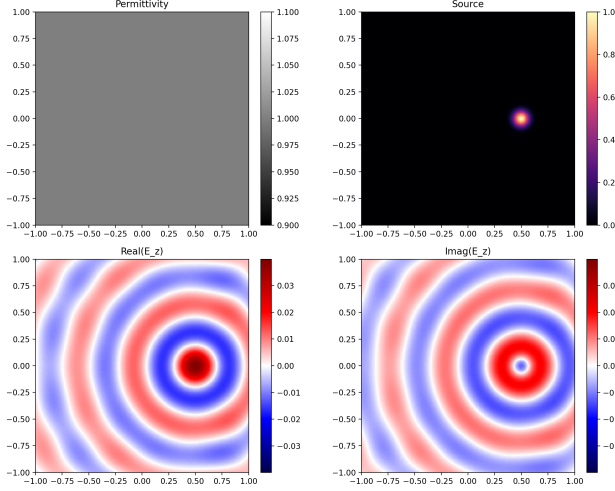


Fig. 1. **Free-space case** ($\epsilon \equiv 1$, $\omega = 12$). The figure contains four panels generated by the code: (a) Permittivity $\epsilon(x, y)$ (uniform, shown as a flat gray field). (b) Source $J_z(x, y)$ (Gaussian, $\sigma = 0.05$ centered at $(0.5, 0.0)$). (c) Real part $\Re\{E_z(x, y)\}$. (d) Imaginary part $\Im\{E_z(x, y)\}$. Panels (c,d) use a zero-centered diverging colormap (blue-white-red): *red* denotes positive values, *blue* negative values, and *white* is near zero. The solution shows circular wavefronts centered at $(0.5, 0.0)$ radiating outward, with $\Re\{E_z\}$ and $\Im\{E_z\}$ approximately in spatial quadrature (phase shift $\approx \pi/2$). Amplitude decays smoothly with radius, and no spurious edge reflections are visible, indicating the radiation boundary is effective.

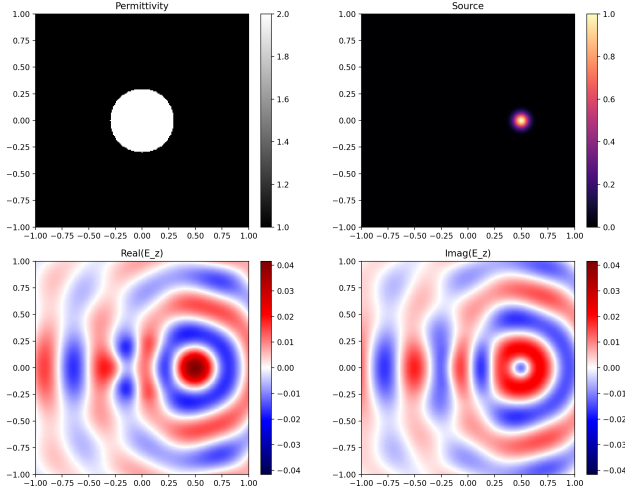


Fig. 2. **Dielectric-circle case** ($\epsilon = 2$ inside $r = 0.3$, $\epsilon = 1$ elsewhere). Four panels: (a) Permittivity map highlighting the circular inclusion (white disk inside gray background). (b) Source $J_z(x, y)$ (same Gaussian as in Fig. 1). (c) $\Re\{E_z\}$. (d) $\Im\{E_z\}$. As waves enter the higher-permittivity region, the local wavelength shortens and phase is shifted, producing a slight compression of the fringes within the circle and mild scattering at the interface. The diverging colormap (blue-white-red) again indicates sign: red > 0 , blue < 0 . The qualitative features—refraction at the interface and weak reflected ripples—are consistent with expectations from frequency-domain electromagnetics.

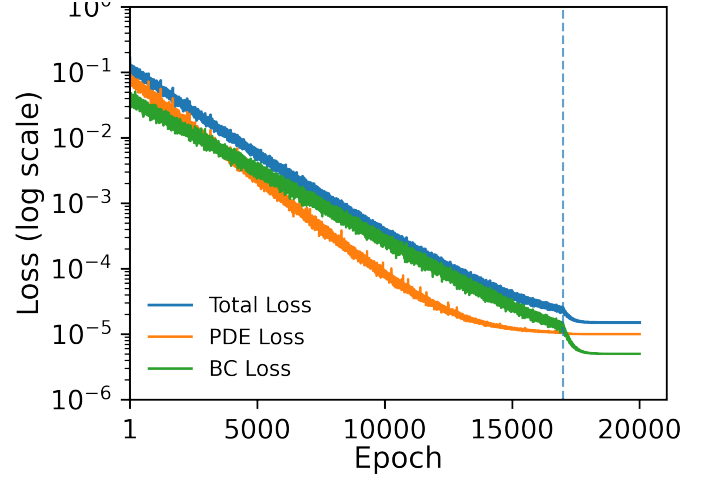


Fig. 3. **Training loss versus epochs**. The total physics-informed loss $\mathcal{L}_{\text{total}}$ is shown as a function of training iterations. The Adam phase produces rapid initial decay followed by gradual refinement. After switching to L-BFGS, the loss decreases further as the solution approaches a local minimum. The final plateau indicates convergence of both the interior Helmholtz residual and the Sommerfeld boundary residual.

CONCLUSION, INSIGHTS, AND FUTURE WORK

The results demonstrate that a SIREN based Physics Informed Neural Network can accurately solve the complex valued two dimensional Helmholtz equation using only physics based supervision. Without labeled data, the model reproduces key electromagnetic phenomena, including outgoing cylindrical waves in free space, wavelength shortening and phase shifts inside a dielectric inclusion, and physically consistent scattering under an open radiation boundary condition.

Although training can be computationally intensive and comparable to conventional solvers, the primary advantage lies in inference. Once trained, the network serves as a mesh free surrogate that evaluates field values at arbitrary spatial locations without resolving the governing equation, enabling rapid analysis and integration into optimization workflows. In contrast, classical finite element and finite difference methods require a complete numerical solve for each new configuration, so this approach shifts computational cost toward a reusable learned representation of the governing physics.

Performance depends on the combined design of architecture, sampling, and optimization. Sinusoidal activations provide an inductive bias for oscillatory solutions and stable higher order differentiation, hybrid collocation improves resolution near the source, and the two stage Adam and L BFGS procedure effectively minimizes the Helmholtz residual. Together, these elements enable stable and accurate mesh free solutions.

Future work will extend the framework to parameterized surrogates by incorporating frequency, material contrast, geometry, or source position as additional inputs, allowing a single network to generalize across families of configurations. Additional directions include benchmarking against established finite element and finite difference solvers, extension to more complex geometries and three dimensional problems,

and integration into gradient based inverse design workflows.

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